

NUMARA P11 Technical Appendix

Schemas, configurations, raw metrics, failure taxonomy, and next experimental protocol for the V0.1-V0.2 harness.

This appendix is intended as reproducibility documentation for a future scientific paper. It documents the current artifact, not a production-ready system.

Artifact Manifest

File	Purpose
graph.json	Primitive-locked node/edge graph
paths.json	Heuristic path field definitions
guards.json	Guard rules and guard floors
config.json	V0.2 stochastic test parameters
tests.yaml	Human-readable test declarations
runner.py	Executable deterministic/stochastic simulator
results.json	Summary metrics from latest run
runs.jsonl	Per-trial logs from V0.2
batch_report.md	Markdown summary report
rollback_manifest.json	File rollback instructions
README.md	Run instructions and boundary statement

Primitive Basis

Index	Primitive
0	P0_observe
1	P1_difference
2	P2_pattern
3	P3_constraint
4	P4_node
5	P5_edge
6	P6_state
7	P7_mode
8	P8_memory
9	P9_test
10	P10_repair
11	P11_equivalence
12	P12_publish

All node vectors are required to remain 13-dimensional over this exact P0-P12 basis. A change in basis order, dimension, or semantic meaning is primitive drift.

Node Inventory

Node	Status	Allowed use	Top primitive notes
task_pressure	Raw	internal_test	P0_observe=0.63, P7_mode=0.39, P6_state=0.32
community_build	Raw	internal_test	P0_observe=0.35, P7_mode=0.35, P12_publish=0.35
grant_output	Raw	internal_test	P11_equivalence=0.45, P2_pattern=0.33, P9_test=0.33
event_output	Raw	internal_test	P11_equivalence=0.44, P12_publish=0.38, P9_test=0.33
durable_process	Candidate	internal_test	P11_equivalence=0.38, P8_memory=0.34, P2_pattern=0.30
community_memory	Candidate	internal_test	P8_memory=0.44, P11_equivalence=0.39, P6_state=0.35
shared_roles	Raw	internal_test	P6_state=0.36, P11_equivalence=0.36, P5_edge=0.31
repair_loop	Raw	internal_test	P10_repair=0.45, P3_constraint=0.32, P6_state=0.32
replicable_model	Raw	internal_test	P9_test=0.36, P11_equivalence=0.36, P12_publish=0.31
local_food_resilience	Raw	internal_test	P0_observe=0.37, P11_equivalence=0.37, P6_state=0.32
metrics_budget	Raw	internal_test	P9_test=0.48, P3_constraint=0.37, P11_equivalence=0.32
human_surface	Raw	internal_test	P12_publish=0.46, P0_observe=0.40, P7_mode=0.35
public_claim_guard	Raw	internal_test	P3_constraint=0.43, P9_test=0.39, P10_repair=0.34
technical_memory	Raw	internal_test	P9_test=0.38, P3_constraint=0.34, P8_memory=0.34
final_output	Raw	internal_test	P12_publish=0.53, P7_mode=0.42, P11_equivalence=0.37
unsupported_output_risk	Raw	internal_test	P3_constraint=0.46, P1_difference=0.41, P9_test=0.41
overtechnical_risk	Raw	internal_test	P9_test=0.43, P12_publish=0.43, P3_constraint=0.37

Edge Inventory

Source	Target	Weight	Type
task_pressure	community_build	0.700	supports
community_build	event_output	0.750	supports
community_build	durable_process	0.720	supports
event_output	durable_process	0.800	translates
grant_output	durable_process	0.820	translates
durable_process	community_memory	0.850	supports
community_memory	repair_loop	0.650	supports
community_memory	replicable_model	0.680	supports
shared_roles	community_memory	0.580	supports
repair_loop	durable_process	0.550	supports
replicable_model	local_food_resilience	0.520	supports
metrics_budget	grant_output	0.500	supports
human_surface	final_output	0.720	supports
durable_process	final_output	0.660	supports
community_memory	final_output	0.620	supports
technical_memory	community_memory	0.540	supports
technical_memory	metrics_budget	0.450	supports
public_claim_guard	unsupported_output_risk	-0.900	inhibits
human_surface	overtechnical_risk	-0.650	inhibits
public_claim_guard	final_output	-0.200	guards
overtechnical_risk	final_output	-0.800	inhibits
unsupported_output_risk	final_output	-0.950	inhibits

Heuristic Path Field

A_coordination_logistics

Bias node	Value
community_build	0.450
shared_roles	0.400
event_output	0.350
final_output	0.200
public_claim_guard [guard]	0.150

B_commons_infra_organizing

Bias node	Value
community_build	0.350
durable_process	0.500
community_memory	0.450
repair_loop	0.350
local_food_resilience	0.300
public_claim_guard [guard]	0.100

C_book_voice_human_surface

Bias node	Value
human_surface	0.550
final_output	0.350
event_output	0.250
community_memory	0.250
public_claim_guard [guard]	0.100

D_exploration_risk_guard

Bias node	Value
durable_process	0.250
repair_loop	0.300
replicable_model	0.250
public_claim_guard [guard]	0.600

E_technical_memory_guarded

Bias node	Value
technical_memory	0.500
metrics_budget	0.300
community_memory	0.300
replicable_model	0.250
public_claim_guard [guard]	0.250
human_surface [guard]	0.250

F_public_followup

Bias node	Value
human_surface	0.400
event_output	0.350
final_output	0.300
shared_roles	0.250
community_memory	0.250

Bias node	Value
public_claim_guard [guard]	0.200

Guard Rules

ID	Condition	Action
G1_basis_lock	all vectors must be 13D over P0-P12	fail test
G2_guard_floor	public follow-up output requires public_claim_guard >= 0.35	suppress final_output if unmet
G3_technical_memory_floor	technical_memory path requires public_claim_guard >= 0.52 or human_surface >= 0.40	flag guard gap
G4_unsupported_output	unsupported_output_risk >= 0.30	suppress final_output
G5_overttechnical_output	overttechnical_risk >= 0.30 and human_surface < 0.40	suppress final_output
G6_no_truth_promotion	internal convergence cannot become external verification	candidate evidence only

V0.2 Configuration

Parameter	Value
harness_id	NUMARA_P11_TEST_HARNESS_V0_2
seed	2026070802
trials_per_path	24
activation_noise	0.055
edge_weight_noise	0.045
edge_dropout	0.08
guard_floor	0.35
technical_memory_guard_floor	0.52
invariant_threshold	0.52
unsafe_risk_max	0.3
strict_invariants_build_day	["event_output", "durable_process", "community_memory"]
false_convergence_tests	{"risk_only_path": ["unsupported_output_risk", "overttechnical_risk"], "expect_invariants": false}
notes	V0.2 adds randomized perturbations, edge dropout, guard adversarial checks, and false-convergence detection.

V0.1 Raw Path Results

A_coordination_logistics	community_memory, durable_process, event_output, final_output, local_food_resilience, repair_loop, replicable_model	0.025	0.000	durable_process=0.978
B_commons_infra_organizing	community_memory, durable_process, event_output, final_output, local_food_resilience, repair_loop, replicable_model	0.017	0.000	durable_process=0.978
C_book_voice_human_surface	community_memory, durable_process, event_output, final_output, local_food_resilience, repair_loop, replicable_model	0.017	0.000	durable_process=0.977
D_exploration_risk_guard	community_memory, durable_process, event_output, final_output, local_food_resilience, repair_loop, replicable_model	0.083	0.000	durable_process=0.977
E_technical_memory_guarded	community_memory, durable_process, event_output, final_output, grant_output, local_food_resilience, public_claim_guard, repair_loop, replicable_model	0.520	0.000	durable_process=0.992
F_public_followup	community_memory, durable_process, event_output, final_output, local_food_resilience, repair_loop, replicable_model	0.033	0.000	durable_process=0.977

V0.2 Derived Metrics

Transfer Path Statistics

Path	Trials	Invariant success	Guard mean	Risk mean	Final mean
A_coordination_logistics	24	66.7%	0.0432	0.0178	0.9459
B_commons_infra_organizing	24	91.7%	0.0397	0.0206	0.9571
C_book_voice_human_surface	24	83.3%	0.0303	0.0141	0.9460
D_exploration_risk_guard	24	79.2%	0.0789	0.0291	0.9450
E_technical_memory_guarded	24	87.5%	0.5200	0.0077	0.9449
F_public_followup	24	83.3%	0.0389	0.0239	0.9516

Adversarial Guard Derived Stats

Metric	Value
Runs	24
Mean guard score	0.600000
Mean unsafe risk	0.042164
Mean final output	0.905725
Final output min/max	0.498348 / 0.965778

The adversarial guard test failed because final_output remained high even when risk-only conditions should suppress publication. Guard score was high, but guard score did not translate into sufficient output suppression.

False-Convergence Derived Stats

Metric	Value
Runs	24
False convergence count	16/24
False convergence rate	66.7%
Mean guard score	0.600000
Mean unsafe risk	0.032184
Mean final output	0.839611

The false-convergence detector failed because risk-only trials still activated the build-day invariant in 16 of 24 runs. This is the strongest evidence of over-convergence in the current graph.

Algorithm Summary

```
For each path stack:
  seed task_pressure
  apply path bias and guard bias
  perturb edges and activations when randomized
  propagate weighted activation for N steps
  apply guard floors and suppression rules
  collect active features above invariant threshold
  score invariant presence, guard behavior, unsafe risk, convergence, and false convergence
```

Known Failure Classes

Failure class	Definition	Repair direction
Primitive drift	Vector basis changes or node role stops matching P0-P12	Basis hash, 13D check, cosine delta sentinels
Unsafe emergent edge	Trap or risk node becomes positive bridge to important output path	Quarantine edge; convert to negative evidence
Guard underactivation	Risk task does not sufficiently activate guard pathway	Raise guard floor; require suppression behavior
Exploratory guard gap	Useful divergent path lacks adequate guard coactivation	Keep candidate-only until clean pass
False convergence	Negative control activates target invariant	Add task gates and context-negative controls

Failure class	Definition	Repair direction
Over-convergence	Known invariant appears too broadly across mismatched contexts	Introduce invariant eligibility rules

V0.3 Protocol

V0.3 should not add more creativity. It should repair task gating. The goal is to reduce false convergence while preserving positive transfer.

V0.3 addition	Purpose
Task gate nodes	Represent whether the task is eligible for event/process/community-memory invariant
Context-negative controls	Risk-only, unrelated, and decoy tasks that should not converge on target invariant
Invariant eligibility rule	Target invariant only counts if task gate is active
Guard-to-output coupling	Guard activation must actually suppress final_output under risk
False-convergence penalty	Any negative-control invariant activation reduces score
Report confidence intervals	Move from point metrics to statistical summaries

Million-Test Scaling Plan

After V0.3 passes negative controls, scale by batches. Do not run one million tests as a monolithic file. Use seed shards, JSONL compression, summary reductions, and reproducible config snapshots.

```
Suggested campaign structure:
seeds: 0..999999
shard_size: 10,000 runs
outputs per shard:
  shard_summary.json
  failures.jsonl.gz
  metrics.csv
global reductions:
  invariant success distribution
  false convergence distribution
  guard pass distribution
  parameter sensitivity heatmap
  failure cluster report
```

Reproducibility Boundary

The harness is reproducible as a local Python artifact. It is not yet independent evidence of cognitive generalization. The next paper-ready step is to define external task suites, negative controls, and human review rubrics before large-scale iteration.